

FSUC0360

## Cadmium iodide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 碘化镉  
Cadmium iodide

**Cat No. :** C/0360/48  
**CAS No** 7790-80-9  
**Molecular Formula** Cd I2

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
Off-white

**Odor**  
Odorless

#### Emergency Overview

Toxic if swallowed. Toxic if inhaled. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure. Very toxic to aquatic life with long lasting effects.

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 3 |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 3 |
| Carcinogenicity                                      | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Acute aquatic toxicity                               | Category 1 |
| Chronic aquatic toxicity                             | Category 1 |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H351 - Suspected of causing cancer

H373 - May cause damage to organs through prolonged or repeated exposure

H410 - Very toxic to aquatic life with long lasting effects

H301 + H331 - Toxic if swallowed or if inhaled

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P311 - Call a POISON CENTER or doctor

P330 - Rinse mouth

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Toxic if swallowed. Toxic if inhaled. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Very toxic to aquatic life with long lasting effects.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component      | CAS No    | Weight % |
|----------------|-----------|----------|
| Cadmium iodide | 7790-80-9 | >95      |

**Note**

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

## Cadmium iodide

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component      | China | Taiwan                      | Thailand | Hong Kong |
|----------------|-------|-----------------------------|----------|-----------|
| Cadmium iodide | -     | TWA: 0.05 mg/m <sup>3</sup> |          | -         |

| Component      | ACGIH TLV                                                                    | OSHA PEL | NIOSH                     | The United Kingdom                                                        | European Union |
|----------------|------------------------------------------------------------------------------|----------|---------------------------|---------------------------------------------------------------------------|----------------|
| Cadmium iodide | TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.002 mg/m <sup>3</sup><br>TWA: 0.01 ppm |          | IDLH: 9 mg/m <sup>3</sup> | STEL: 0.075 mg/m <sup>3</sup> 15 min<br>TWA: 0.025 mg/m <sup>3</sup> 8 hr |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** In case of insufficient ventilation, wear suitable respiratory equipment

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

## Cadmium iodide

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When RPE is used a face piece Fit Test should be conducted

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Appearance                              | Off-white                |                                   |
| Physical State                          | Solid                    |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | 5                        | 50 g/l aq.sol                     |
| Melting Point/Range                     | 388 °C / 730.4 °F        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 787 °C / 1448.6 °F       |                                   |
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No information available |                                   |
| Vapor Density                           | No information available | (Air = 1.0)                       |
| Specific Gravity / Density              | 5.67                     |                                   |
| Bulk Density                            | No data available        |                                   |
| Water Solubility                        | 798 g/L (20°C)           |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |
| Molecular Formula                       | Cd I2                    |                                   |
| Molecular Weight                        | 366.2                    |                                   |

**SECTION 10. STABILITY AND REACTIVITY**

|                          |                                          |
|--------------------------|------------------------------------------|
| Stability                | Stable under normal conditions.          |
| Hazardous Reactions      | None under normal processing.            |
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Conditions to Avoid      | Incompatible products.                   |
| Materials to avoid       | Strong oxidizing agents. Acids.          |

Hazardous Decomposition Products Hydrogen iodide.

**SECTION 11. TOXICOLOGICAL INFORMATION**

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|-----------|-------------|-----------------|
|-----------|-----------|-------------|-----------------|

**SAFETY DATA SHEET****Cadmium iodide**

|                |                          |  |  |
|----------------|--------------------------|--|--|
| Cadmium iodide | LD50 = 222 mg/kg ( Rat ) |  |  |
|----------------|--------------------------|--|--|

|                                            |                                                           |
|--------------------------------------------|-----------------------------------------------------------|
| (b) skin corrosion/irritation;             | No data available                                         |
| (c) serious eye damage/irritation;         | No data available                                         |
| (d) respiratory or skin sensitization;     |                                                           |
| Respiratory                                | No data available                                         |
| Skin                                       | No data available                                         |
| (e) germ cell mutagenicity;                | No data available                                         |
| (f) carcinogenicity;                       | No data available                                         |
|                                            | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;                 | No data available                                         |
| (h) STOT-single exposure;                  | No data available                                         |
| (i) STOT-repeated exposure;                | No data available                                         |
| Target Organs                              | Lungs, Kidney.                                            |
| (j) aspiration hazard;                     | No data available                                         |
| Other Adverse Effects                      | See actual entry in RTECS for complete information        |
| Symptoms / effects, both acute and delayed | No information available                                  |

**SECTION 12. ECOLOGICAL INFORMATION**

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Ecotoxicity effects             | Do not empty into drains.                                                 |
| Persistence and Degradability   | No information available                                                  |
| Bioaccumulative Potential       | No information available                                                  |
| Mobility in soil                | No information available                                                  |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                            |                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------|
| Waste from Residues/Unused | Chemical waste generators must determine whether a discarded chemical is classified as a |
|----------------------------|------------------------------------------------------------------------------------------|

## Cadmium iodide

|                               |                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>               | hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
| <b>Contaminated Packaging</b> | Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.                         |
| <b>Other Information</b>      | Waste codes should be assigned by the user based on the application for which the product was used.                                |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                                |                  |
|--------------------------------|------------------|
| <b>UN-No</b>                   | UN2570           |
| <b>Proper Shipping Name</b>    | CADMIUM COMPOUND |
| <b>Technical Shipping Name</b> | Cadmium iodide   |
| <b>Hazard Class</b>            | 6.1              |
| <b>Packing Group</b>           | III              |

IMDG/IMO

|                                |                  |
|--------------------------------|------------------|
| <b>UN-No</b>                   | UN2570           |
| <b>Proper Shipping Name</b>    | CADMIUM COMPOUND |
| <b>Technical Shipping Name</b> | Cadmium iodide   |
| <b>Hazard Class</b>            | 6.1              |
| <b>Packing Group</b>           | III              |

IATA

|                                |                  |
|--------------------------------|------------------|
| <b>UN-No</b>                   | UN2570           |
| <b>Proper Shipping Name</b>    | CADMIUM COMPOUND |
| <b>Technical Shipping Name</b> | Cadmium iodide   |
| <b>Hazard Class</b>            | 6.1              |
| <b>Packing Group</b>           | III              |

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component      | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Cadmium iodide | -                                                   | X                                       | X    | X     | 232-223-6 | X    | X   | X     | X    | X    | X    | KE-04412 |

**Note**

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

**National Regulations**

## SECTION 16. OTHER INFORMATION

**Creation Date** 25-Oct-2010  
**Revision Date** 04-Apr-2024  
**Revision Summary** Not applicable.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**